COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
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ASSIGNMENT 9.1 **Binary Search Tree**

1. PROCEDURE OUTPUT

What procedures did you perform?

1. What is a binary search tree?

- A Binary Search Tree (BST) is a type of data structure used to organize and store values. It starts with a root node, and each node can have a maximum of two children. Values smaller than the node are placed on the left, while values larger than the node are placed on the right. Because the values are arranged in this order, a BST makes searching, inserting, and deleting data easier and faster.

2. Where can binary search trees be used?

- A Binary Search Tree (BST) is used to search data that is dynamic. Meaning, it is always searched, updated, inserted and sorted because normal search algorithms are slower on average. It balances memory efficiency with speed. It is usually used in memory databases and caches. Main memory databases use BST variations for rapid lookups, range queries and memory indexing without heavy disk overhead. It can also be used to track minimum and maximum elements simultaneously because trees have traversal patterns that can be used to efficiently track every element of the group. In our daily technology, trees are used in contact lists, directories and in searching since new entries happen constantly while maintaining the alphabetical order for easier navigation and fast searching. It is also used in internet IP addresses mapping to quickly match destination addresses to the local address.

3. What is a tree traversal?

- A tree traversal is the process of accessing or processing each node exactly once inside a tree data structure. Unlike traversal in linear data structures, which only have one logical way of traversing through the list by traversing from the first element or node to the last one, a tree traversal supports various traversal methods since trees have nodes that lead to different directions, such as pre-order, in-order, and post-order traversal methods. It has two classifications of traversal methods, which are depth and height traversal. Depth traversal visits the bottom leaf nodes, while height traversal visits the nodes at one level before heading to the next available level.

4. Differentiate a post-order and pre-order traversal give examples.

- Pre-order traversal visits the root or main value first, then the left side, and then the right side. Post-order traversal visits the left or lowest value first, then the right side, and the root or main value last. For example, if the tree has A at the top, B on the left, and C on the right, pre-order is A-B-C, while post-order is B-C-A. The main difference is that pre-order starts with the root, while post-order ends with the root.

5. What is a parse tree and its use?

- A parse tree, or derivation tree, is a tree associated with a derivation of a string from one grammar. The root of the tree represents the entire input string, internal nodes represent grammar rules, and leaves represent tokens of the input. The use of parse trees is that they represent the context-free structure of some input, they show the derivation history and grouping of the terms. This helps compilers and interpreters identify valid syntax and build target code, and process data for language processors, by analyzing the grammatical structure.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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